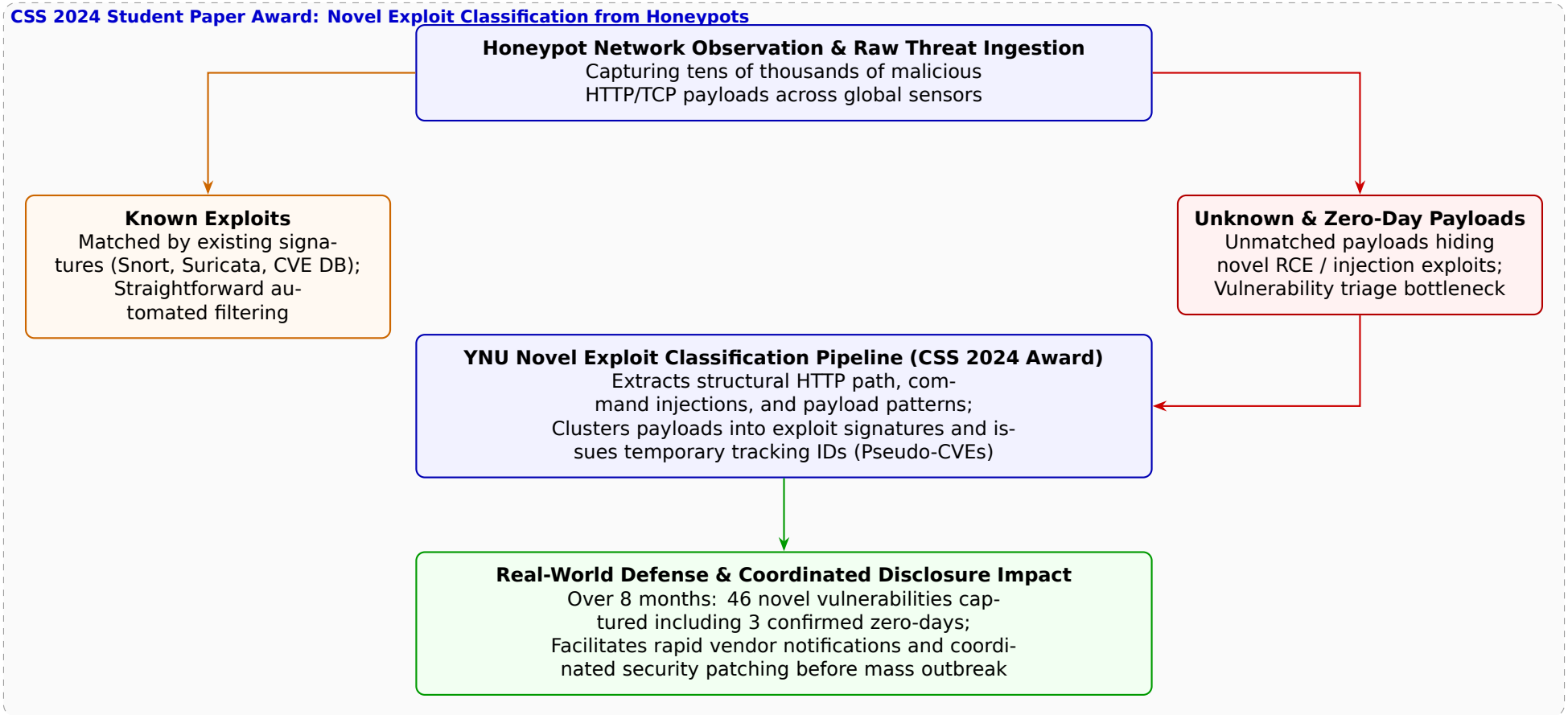
CSS 2024 Student Paper Award: Novel Exploit Classification from Honeypots
Honeypot Network Observation & Raw Threat Ingestion
Capturing tens of thousands of malicious HTTP/TCP payloads across global sensors
Known Exploits
Matched by existing signatures (Snort, Suricata, CVE DB); Straightforward automated filtering
Unknown & Zero-Day Payloads
Unmatched payloads hiding novel RCE / injection exploits; Vulnerability triage bottleneck
YNU Novel Exploit Classification Pipeline (CSS 2024 Award)
Extracts structural HTTP path, command injections, and payload patterns; Clusters payloads into exploit signatures and issues temporary tracking IDs (Pseudo-CVEs)
Real-World Defense & Coordinated Disclosure Impact
Over 8 months: 46 novel vulnerabilities captured including 3 confirmed zero-days; Facilitates rapid vendor notifications and coordinated security patching before mass outbreak